

## 005 · One Function, Many Coasters — Parameters — English Worksheet (Beginner)

**Topic:** One Function, Many Coasters — Parameters · **Course:** Theme Park Creator · **Level:** Beginner ·

**Time:** about 30 minutes

This lesson you build a roller coaster with a **function**. You give it **parameters** — numbers it uses inside — and change them to build a long or short coaster.

These are the blocks you use this lesson:

```
rollerCoasterBuilder.add_straight_line(10)
rollerCoasterBuilder.add_ramp(RcbVerticalDirection.UP, 10)
rollerCoasterBuilder.add_ramp(RcbVerticalDirection.DOWN, 10)
```

A direction is **UP** or **DOWN**. The number after it is how many blocks the ramp goes.

### 1 · Predict 🧠

Read the code. Circle your answer.

```
def build_hill(line_len):
    rollerCoasterBuilder.add_straight_line(line_len)
    rollerCoasterBuilder.add_ramp(RcbVerticalDirection.UP, line_len)

build_hill(6)
```

**The straight line will be 6 blocks long. Circle one:** yes · no

**Why 6? Circle one:** we called **build\_hill(6)** · we called **build\_hill(10)**

```
def build_hill(line_len):
    rollerCoasterBuilder.add_straight_line(line_len)
    rollerCoasterBuilder.add_ramp(RcbVerticalDirection.UP, line_len)
```

**There is no call after the definition. Does anything get built? Circle one:** yes · no

## 2 · Find the Difference 🐉

Clean code first, then a broken version. Circle the answer.

**Pair A** — The function should be **called** so the hill gets built.

```
# clean
def build_hill(line_len):
    rollerCoasterBuilder.add_straight_line(line_len)
    rollerCoasterBuilder.add_ramp(RcbVerticalDirection.UP, line_len)

build_hill(6)
```

```
# buggy
def build_hill(line_len):
    rollerCoasterBuilder.add_straight_line(line_len)
    rollerCoasterBuilder.add_ramp(RcbVerticalDirection.UP, line_len)
```

**What is wrong? Circle one:** no call after `def` · wrong number · wrong color

**Pair B** — The function should use the **parameter** `line_len`, not a fixed number.

```
# clean
def build_hill(line_len):
    rollerCoasterBuilder.add_straight_line(line_len)
```

```
# buggy
def build_hill(line_len):
    rollerCoasterBuilder.add_straight_line(10)
```

**What is wrong? Circle one:** it uses `10`, not `line_len` · it uses `line_len`, not `10` · nothing

### 3 · Fill the Gap

This coaster goes **up** then **down** so the cart lands safely. One word is missing.

```
def up_and_down(ramp_len):  
    rollerCoasterBuilder.add_ramp(RcbVerticalDirection.UP, ramp_len)  
    rollerCoasterBuilder.add_ramp(RcbVerticalDirection.____, ramp_len)  
  
up_and_down(10)
```

Which word is missing? Circle one: DOWN · UP · LEFT

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## 4 · Show Your Work 📷 🎥

Switch to your homework world. Write one `build_hill` function with a `line_len` parameter and a ramp up, then call it two times with different numbers.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

**1 · Your code.** Scroll through it. Say what each part does.

**2 · Your build.** Point the camera. Name the parts.

Say these out loud, filling in the blanks:

Today I built \_\_\_\_\_.

I built it using this code: \_\_\_\_\_.

In this code I used \_\_\_\_\_.

The hardest part was \_\_\_\_\_.

That part was hard because \_\_\_\_\_.

The most fun part was \_\_\_\_\_.

Something new I learned was \_\_\_\_\_.

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